

NUMERICAL ANALYSIS

PROBLEM SET 3

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We begin by noting that all of the files referenced in this document are available at:
<https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet3>

1. PROBLEM 1

We'll be approximating $\sqrt{2}$ though this problem

1.a. Our goal is to find a polynomial interpolant $p(x)$ using the Lagrange form given that we have three points in the form $x, f(x)$ as $\{(-1, 1/2), (0, 1), (1, 2)\}$. So we let $f(x) = 2^x$ such that $f(1/2) = \sqrt{2}$. So our polynomial begins as

$$p(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x),$$

with $y_i = f(x_i)$ and $L_i(x)$ being the i^{th} Lagrange polynomial. Computing the polynomials follows as

$$\begin{aligned} L_{2,0}(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 0)(x - 1)}{(-1 - 0)(-1 - 1)} = \frac{x(x - 1)}{2} \\ L_{2,1}(x) &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x + 1)(x - 1)}{(0 + 1)(0 - 1)} = 1 - x^2 \\ L_{2,2}(x) &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x + 1)(x - 0)}{(1 + 1)(1 - 0)} = \frac{x(x + 1)}{2} \end{aligned}$$

1.b. Putting these together we have the following polynomial:

$$\begin{aligned} p(x) &= \frac{1}{2} \cdot \frac{x(x - 1)}{2} + 1 \cdot (1 - x^2) + 2 \cdot \frac{x(x + 1)}{2} \\ &= \frac{x^2}{4} - \frac{x}{4} + 1 - x^2 + x^2 + x \\ &= \frac{x^2}{4} + \frac{3x}{4} + 1 \end{aligned}$$

Pluggin in $x = 1/2$ we get the following

$$\begin{aligned} p\left(\frac{1}{2}\right) &= \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} + 1 \\ &= \frac{1}{16} + \frac{3}{8} + 1 \\ &= \frac{1}{16} + \frac{6}{16} + \frac{16}{16} \\ &= \frac{23}{16} \approx \sqrt{2}. \end{aligned}$$

1.c. For the error bound we want to compute the following

$$|f(x) - p(x)| = \frac{f'''(\xi)}{3!}(x - x_0)(x - x_1)(x - x_2),$$

where $\xi \in [-1, 1]$. We compute the third derivative of $f(x)$ by setting $f(x) = \ln(x) = e^{x \ln(2)}$. Clearly then we have

$$f'''(x) = (\ln(2))^3 e^{x \ln(2)} = (\ln(2))^3 2^x.$$

We want to have ξ such that $|f'''(\xi)|$ is largest so that everything else is beneath that as our upper bound. Because $f(x) = 2^x$ is an increasing exponential, the maximum of the third derivative will necessarily be the right hand end point. So we set $\xi = 1$ giving us $f'''(\xi) = 2(\ln(2))^3$. So

$$|f(x) - p(x)| = \frac{f'''(\xi)}{3!}(x - x_0)(x - x_1)(x - x_2) = \frac{2(\ln(2))^3}{6}(x + 1)(x)(x - 1)$$

And we want to know the error at $x = 1/2$ which comes out at

$$\begin{aligned} \left| f\left(\frac{1}{2}\right) - p\left(\frac{1}{2}\right) \right| &= \frac{2(\ln(2))^3}{6} \left(\frac{1}{2} + 1 \right) \cdot \frac{1}{2} \cdot \left(\frac{1}{2} - 1 \right) \\ &= \frac{3 \ln(2)^3}{24} \\ &= \frac{\ln(2)^3}{8} = 0.0416 \end{aligned}$$

2. PROBLEM 2

Our goal here is to approximate $\ln(2)$ with piecewise polynomials.

2.a. Since each piece has 3 coefficients and there are $n + 1$ points there are n pieces to the polynomial. So we have $3n$ coefficients to account for. We will refer to our interpolation function at $s(x)$, with the piceces being $s_i(x)$.

We begin by interpolating the endpoints of the segments giving us $s_i(x_i) = y_i$ and $s_i(x_{i+1}) = y_{i+1}$. This gives us our first $2n$ constraints.

Next we approach the first derivatives of the pieces for which we need equality for continuity on the inner $n - 1$ points or knots. So we must have that

$$s'_i(x_{i+1}) = s'_{i+1}(x_{i+1}),$$

which gives us another $n - 1$ conditions. So we are missing 1 condition to reach our $3n$ requisite conditions to be able to solve for

The piece that we are missing is one of the boundary conditions. For example we could use $s_0(x_0)$ and fix that or use $s_{n-1}(x_n)$. Regardless we need this one more condition to get all of our constants.

2.b. We were instructed to use the dataset from problem 1, $\{(-1, 1/2), (0, 1), (1, 2)\}$ to solve this problem. This yields the intervals $I_0 = [-1, 0]$ and $I_1 = [0, 1]$. The interpolating polynomials that we are looking to find the coefficients for follow

$$\begin{aligned} s_0(x) &= a_0 + b_0(x + 1) + c_0(x + 1)^2 \\ s_1(x) &= a_1 + b_1x + c_1x^2. \end{aligned}$$

We begin by using the interpolation conditions such that $s_0(-1) = 1/2 \Rightarrow a_0 = 1/2$, $s_0(0) = 1 = 1/2 + b_0 + c_0$ and $s'_0(-1) = b_0 + 2c_0(-1 + 1) = 0$ we have that $b_0 = 0$. Thus $c_0 = 1/2$, which gives us

$$s_0(x) = \frac{1}{2} + \frac{1}{2}(x + 1)^2$$

Notice that $s_1(0) = 1 \Rightarrow a_1 = 1$. Next we use derivative continuity at $x = 0$ to find the other coefficients. We have that $s'_1(x) = b_1 + 2c_1x$, so $s'_0(0) = s'_1(0) = b_1 = 1$. Lastly, we use our boundary condition $s_1(1) = 2 = 1 + 1 + c_1(1) = 2$ to get $c_1 = 0$. And we have

$$s_1(x) = 1 + x.$$

So

$$s(x) = \begin{cases} \frac{1}{2} + \frac{1}{2}(x + 1)^2 & -1 \leq x < 0 \\ 1 + x & 0 \leq x \leq 1 \end{cases}$$

2.c. We begin by noticing that $s'_0(0) = 1$ and $s'_1(0) = 1$ by the continuity of $s'(x)$. Now again we are using the function $f(x) = 2^x$ such that $f'(x) = \ln(2)2^x$ and $f'(0) = \ln(2)$, so we use $s'(0) = 1 \approx \ln(2)$. We did a calculation on a computer just to test the error here and we got $\ln(2) = 0.69315$, so we are quite off

2.d. Skip

3. PROBLEM 3